KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 3

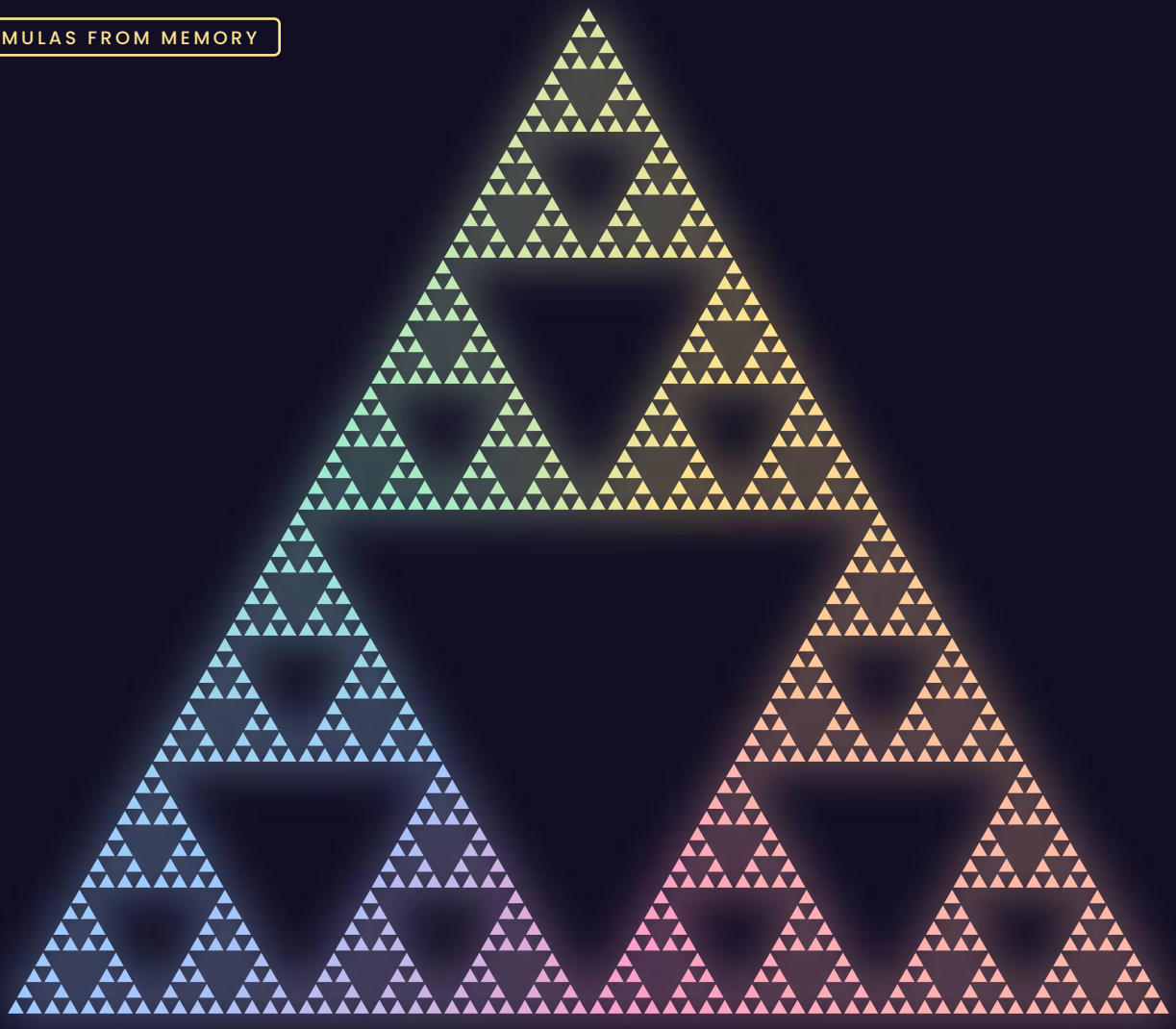
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 10

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS FROM MEMORY



NAME _____

CLASS _____

SIERPIŃSKI TRIANGLE
three copies of itself, each half the size

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

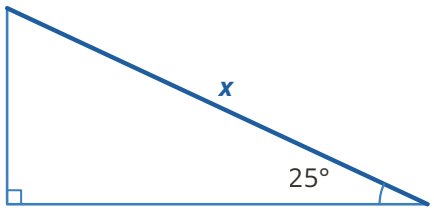
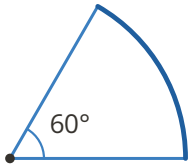
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Ruby earns a salary of \$91 000 a year. Find Ruby's weekly pay.	FORMULA SUBSTITUTE ANSWER
4 Write as a single power. $(5^2)^3$	FORMULA SUBSTITUTE ANSWER
5 Find the value of y when $x = -4$. $y = -x + 6$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

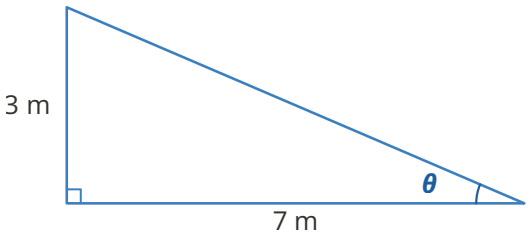
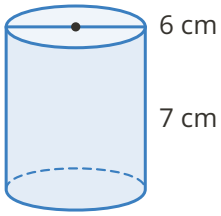
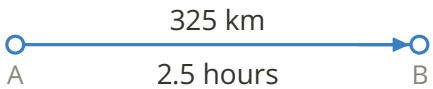
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the percentage loss. Cost price \$125 Selling price \$115	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 A train travels from A to B. Find the average speed. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the mean of these scores, correct to 1 decimal place. Score: 2, 3, 4, 5, 6 Frequency: 2, 5, 9, 6, 3	FORMULA _____ SUBSTITUTE _____ ANSWER _____

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

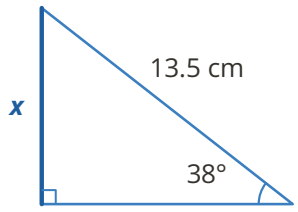
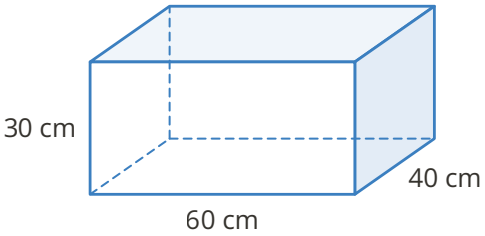
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold? 	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $3500 Interest rate: 2.8% per year Time: 4½ years	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 0.9	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the team does not win. $P(\text{win}) = 0.35$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

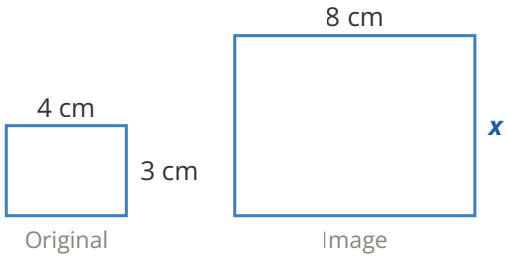
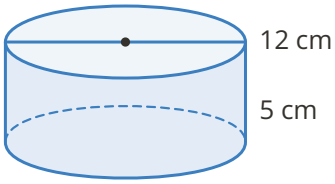
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the price before the increase. Increase: 5% New price: $81.90	FORMULA SUBSTITUTE ANSWER
4 Share $630 in the ratio 2 : 3 : 4. Give all three shares.	FORMULA SUBSTITUTE ANSWER
5 A fair coin is tossed 50 times. How many heads would you expect?	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

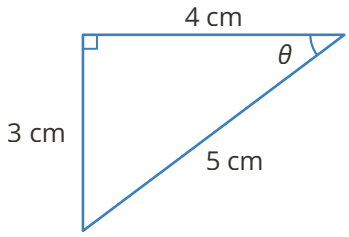
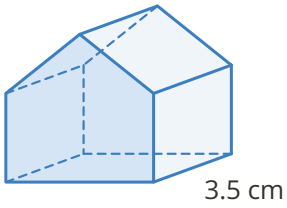
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\cos\theta$ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 The shaded face has an area of 20 cm². Find the volume of the prism.  	FORMULA SUBSTITUTE ANSWER
3 Find Kai's total pay for the week. Hourly rate: $26 Normal hours: 38 hours Time-and-a-half: 6 hours	FORMULA SUBSTITUTE ANSWER
4 Write with a positive index. 7^{-3}	FORMULA SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is white. Colour: Red, White Marbles: 5, 3	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

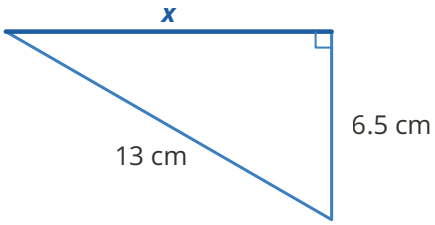
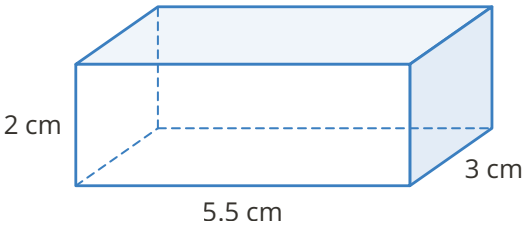
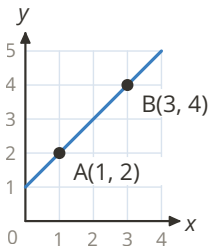
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this prism. 	FORMULA SUBSTITUTE ANSWER
3 Find Omar's total holiday pay. Weekly pay: $1500 Holiday: 4 weeks Leave loading: 17.5%	FORMULA SUBSTITUTE ANSWER
4 Find F when $C = -5$. $F = 1.8C + 32$	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

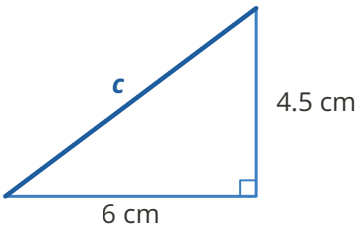
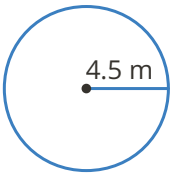
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the new weekly wage. Weekly wage \$900 Pay rise 4%	FORMULA SUBSTITUTE ANSWER
4 Share \$360 in the ratio 2 : 3 : 4. Give all three shares.	FORMULA SUBSTITUTE ANSWER
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{2}{5}$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

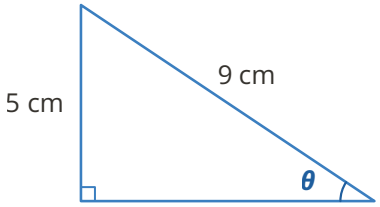
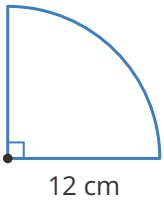
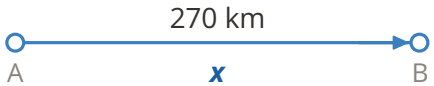
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find Tom's commission. Value of sales: $685 000 Commission rate: 2.5%	FORMULA SUBSTITUTE ANSWER
4 A train travels from A to B at an average speed of 60 km/h. Find x. 	FORMULA SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = -4x + 1$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

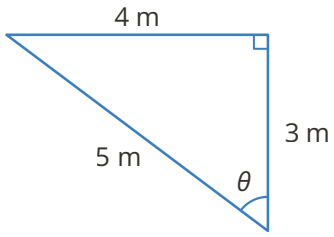
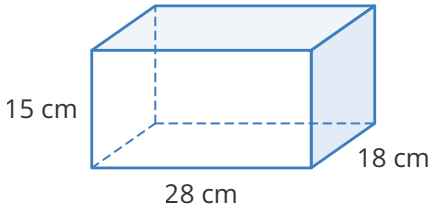
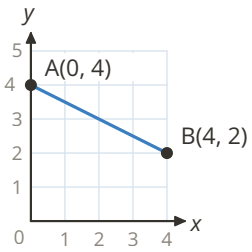
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\tan\theta$ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold, correct to 1 decimal place?  	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $2000 Interest rate: 4% per year Time: 3 years	FORMULA SUBSTITUTE ANSWER
4 Evaluate. 9^0	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

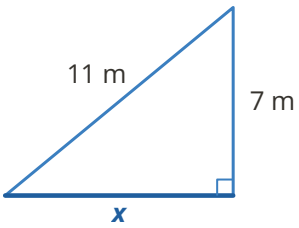
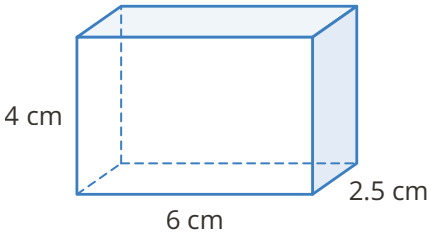
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA SUBSTITUTE ANSWER
3 Find Nina's total holiday pay. Weekly pay: $1263.50 Holiday: 4 weeks Leave loading: 17.5%	FORMULA SUBSTITUTE ANSWER
4 Find h when $A = 24$ and $b = 8$. $A = \frac{1}{2}bh$	FORMULA SUBSTITUTE ANSWER
5 A six came up 15 times in 60 rolls. Find the relative frequency of a six.	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

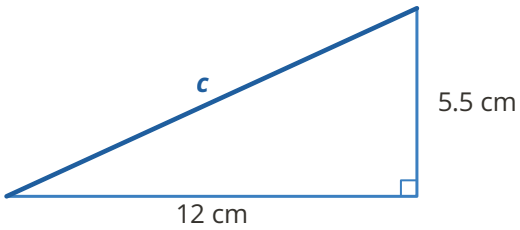
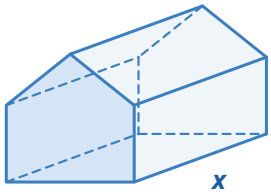
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find c, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 The shaded face has an area of 20 m^2 and the volume is 140 m^3. Find x.  	FORMULA SUBSTITUTE ANSWER
3 Find Ruby's total pay for the week. Weekly base wage: $380 Commission: 5% of sales Sales this week: $7860	FORMULA SUBSTITUTE ANSWER
4 Write as a basic numeral. 1.6×10^{-5}	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these masses. kg: 9.5, 8, 10.5, 12, 7	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

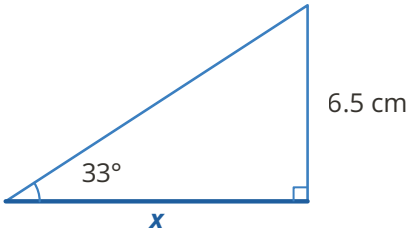
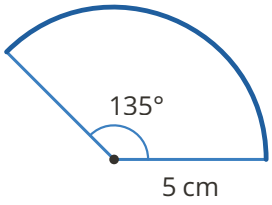
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Tom earns a salary of $74 250 a year. Find Tom's fortnightly pay, to the nearest cent.	FORMULA SUBSTITUTE ANSWER
4 Simplify. $2a^3 \times 7a^4$	FORMULA SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is green. Colour: Red, Blue, Green Marbles: 3, 4, 5	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

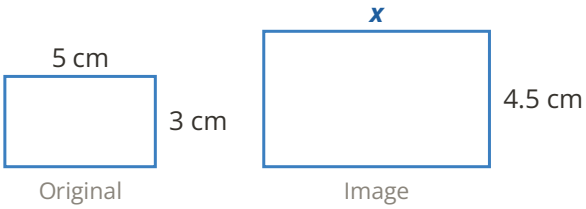
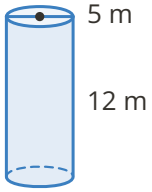
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the rent before the increase. Increase 7% New rent $535	FORMULA SUBSTITUTE ANSWER
4 Share $150 in the ratio 2 : 3. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the player does not win. $P(\text{win}) = 0.47$	FORMULA SUBSTITUTE ANSWER

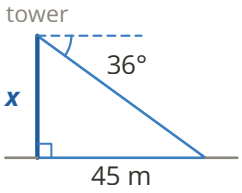
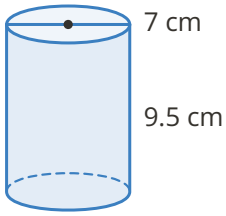
KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Week 5 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 The angle of depression is 36°. Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the sale price. Original price: $89.95 Discount: 20%	FORMULA SUBSTITUTE ANSWER
4 Find P when $l = 8$ and $w = 5$. $P = 2(l + w)$	FORMULA SUBSTITUTE ANSWER
5 Heads came up 7 times in 40 tosses. Find the relative frequency of heads.	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

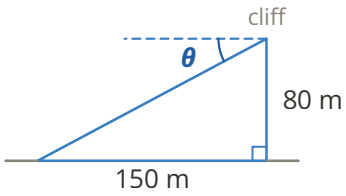
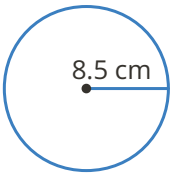
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the angle of depression, θ, correct to the nearest degree. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Hamish earns \$23.60 an hour and works 38 hours. How much does Hamish earn?	FORMULA SUBSTITUTE ANSWER
4 Evaluate. 10^{-3}	FORMULA SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is green. Colour: Red, Blue, Green Marbles: 4, 2, 6	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

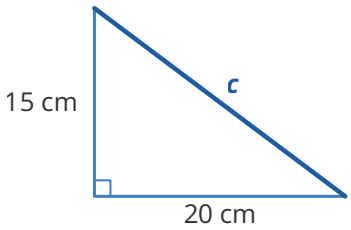
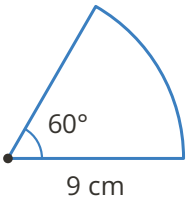
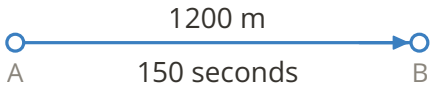
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the percentage profit. Cost price \$36.40 Selling price \$45.50	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 A runner travels from A to B. Find the average speed. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the mean of these scores, correct to 1 decimal place. Score: 5, 6, 7, 8, 9 Frequency: 3, 4, 6, 5, 4	FORMULA _____ SUBSTITUTE _____ ANSWER _____

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

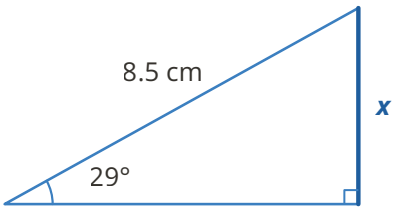
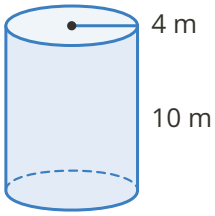
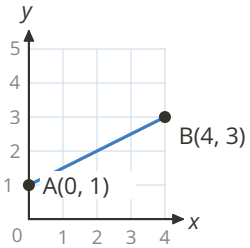
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Aisha earns a salary of $82 750 a year. Find Aisha's weekly pay, to the nearest cent.	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 0.0062	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

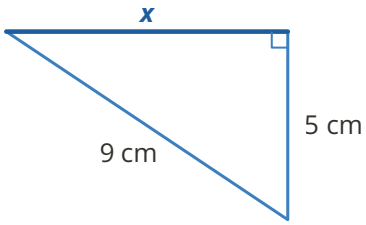
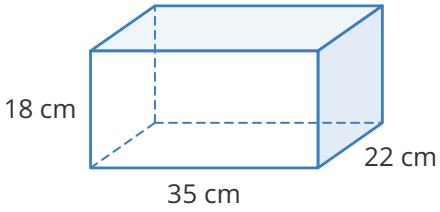
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold, correct to 1 decimal place? 	FORMULA SUBSTITUTE ANSWER
3 Find Jo's total holiday pay. Weekly pay: $1200 Holiday: 4 weeks Leave loading: 17.5%	FORMULA SUBSTITUTE ANSWER
4 Simplify. $15m^8 \div 5m^3$	FORMULA SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = 6 - 2x$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

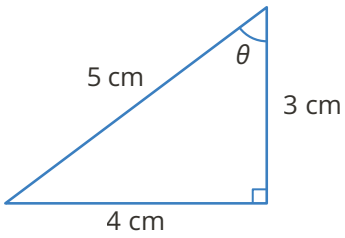
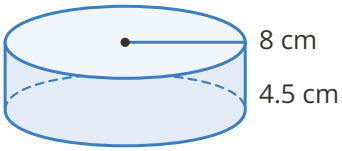
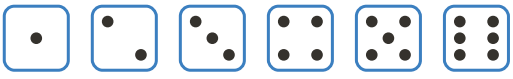
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\sin\theta$ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the original price of the backpack. Discount: 30% Sale price: $56	FORMULA SUBSTITUTE ANSWER
4 Find t when $v = 20$, $u = 4$ and $a = 2.5$. $v = u + at$	FORMULA SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number less than 3. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

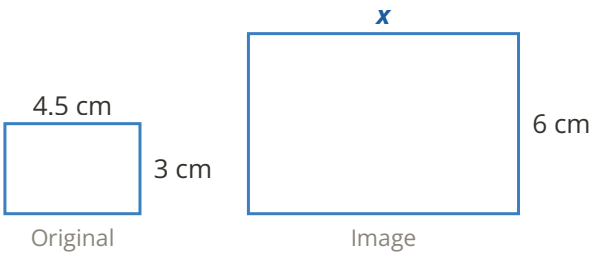
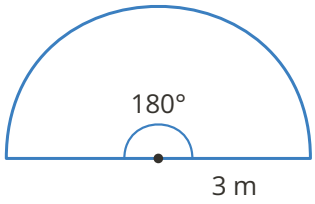
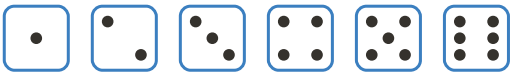
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find Tom's total pay for the week. Hourly rate: $31.40 Normal hours: 35 hours Double time: 6 hours	FORMULA SUBSTITUTE ANSWER
4 Evaluate. 3×2^{-2}	FORMULA SUBSTITUTE ANSWER
5 A fair die is rolled 300 times. How many times would you expect a 4? 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

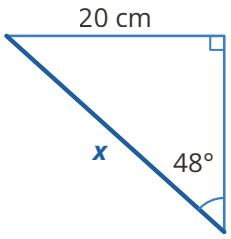
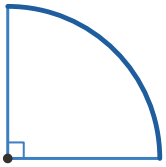
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $15 000 Interest rate: 5.5% per year Time: 2 years	FORMULA SUBSTITUTE ANSWER
4 Simplify. $(3a^2)^3$	FORMULA SUBSTITUTE ANSWER
5 Find the value of y when $x = 2.5$. $y = 4x - 7$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

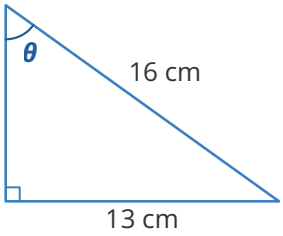
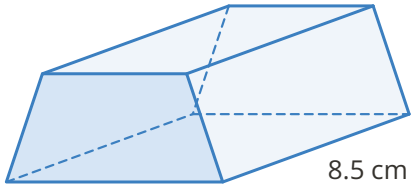
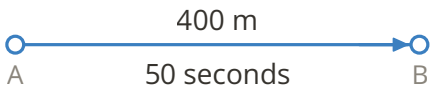
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA SUBSTITUTE ANSWER
2 The shaded face has an area of 15 cm^2. Find the volume of the prism.  	FORMULA SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price: $75 Selling price: $60	FORMULA SUBSTITUTE ANSWER
4 A runner travels from A to B. Find the average speed.  	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the student does not pass. $P(\text{pass}) = 0.82$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

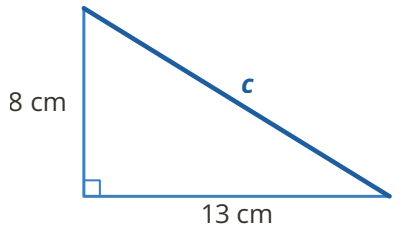
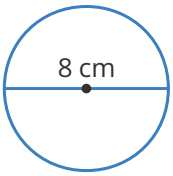
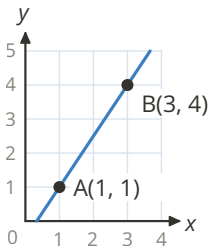
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find c, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the sale price. Original price: $400 Discount: 15%	FORMULA SUBSTITUTE ANSWER
4 Share $360 in the ratio 7 : 5. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

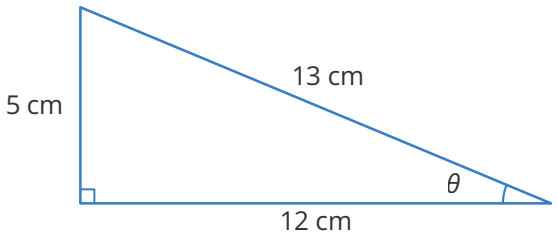
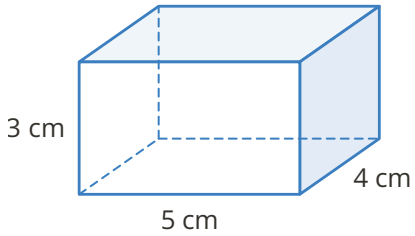
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\sin\theta$ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA SUBSTITUTE ANSWER
3 Find Noah's total pay for the week. Weekly base wage: $450 Commission: 3% of sales Sales this week: $12 500	FORMULA SUBSTITUTE ANSWER
4 Write as a basic numeral. 3.08×10^{-2}	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these marks. Marks: 65, 72, 58, 80, 70	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

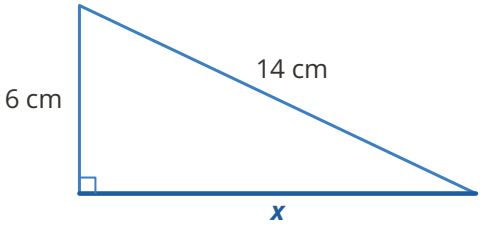
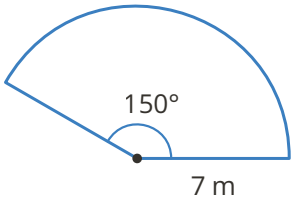
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest on this loan. Amount borrowed: $6000 Interest rate: 4% per year Time: 18 months	FORMULA SUBSTITUTE ANSWER
4 Simplify. $(5y)^0$	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{5}{12}$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

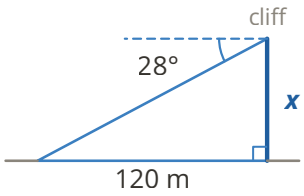
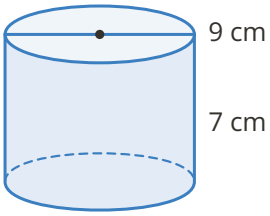
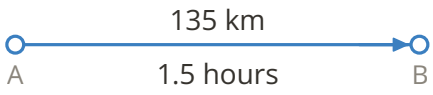
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 The angle of depression is 28°. Find x, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the original price of the jacket. Discount 12.5% Sale price $70	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 A car travels from A to B. Find the average speed. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 A letter is chosen at random from this word. Find $P(S)$. SUCCESS	FORMULA _____ SUBSTITUTE _____ ANSWER _____

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

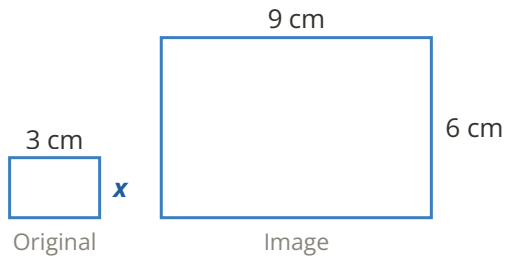
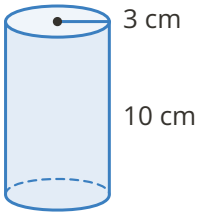
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price \$40 Selling price \$50	FORMULA SUBSTITUTE ANSWER
4 Simplify. $18y^6 \div 6y$	FORMULA SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = -2x + 7$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

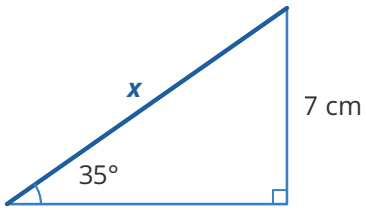
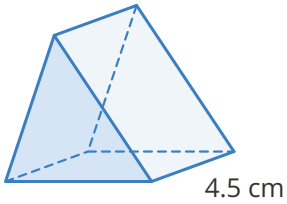
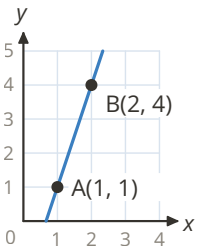
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 The shaded face has an area of 18 cm^2. Find the volume of the prism. 	FORMULA SUBSTITUTE ANSWER
3 Find Ben's total holiday pay. Weekly pay: $985 Holiday: 4 weeks Leave loading: 17.5%	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 150 000	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

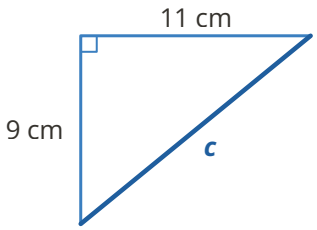
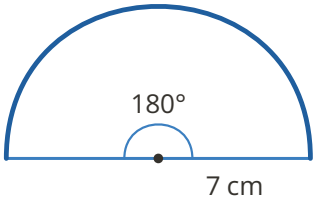
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find c, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Ava earns $25 an hour and works 30 hours. How much does Ava earn?	FORMULA SUBSTITUTE ANSWER
4 Find d when $C = 17$. $C = 5 + 2.4d$	FORMULA SUBSTITUTE ANSWER
5 A spinner was spun 100 times. Find the relative frequency of green. Colour: Red, Blue, Green Frequency: 23, 41, 36	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

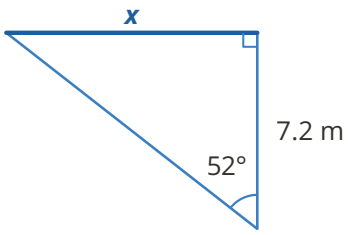
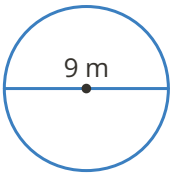
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Leo earns a salary of \$95 300 a year. Find Leo's fortnightly pay, to the nearest cent.	FORMULA SUBSTITUTE ANSWER
4 Share \$200 in the ratio 1 : 4. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 8, 11, 7, 14, 10	FORMULA SUBSTITUTE ANSWER